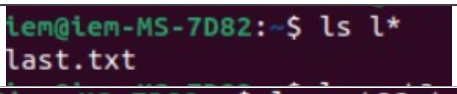
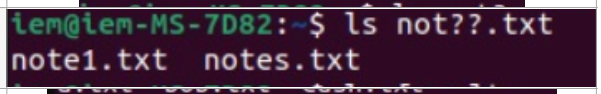
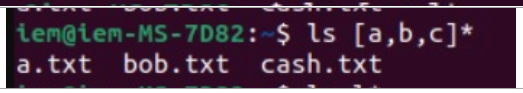
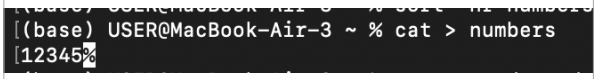
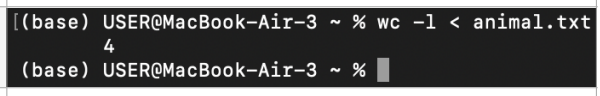
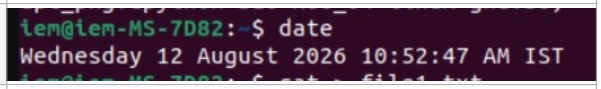
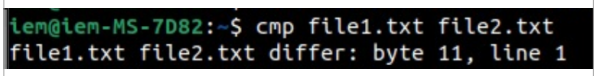
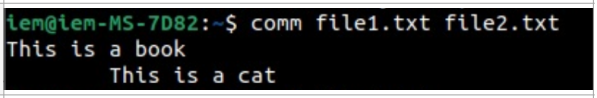
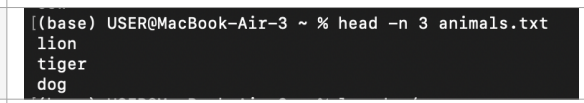
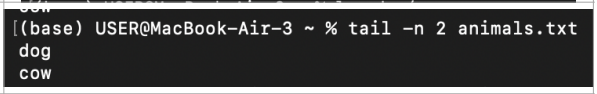
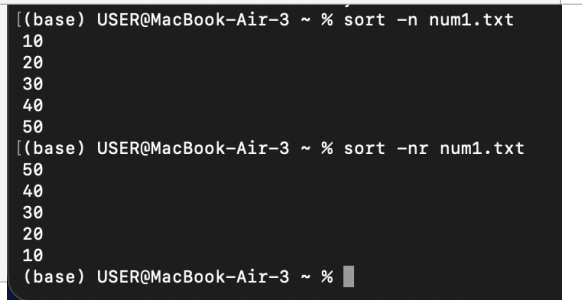
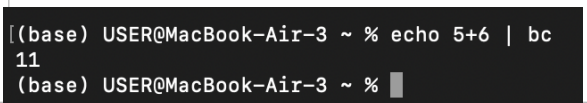
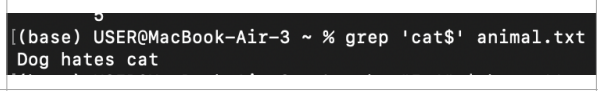
ASSIGNMENT -2

0. For each command, give a brief description of what it does and two examples of how it can be used

Command	Description	Syntax	
*	Matches **zero or more characters** in filenames.	`$ ls l*`	
?	Matches **exactly one character** in a filename.	`$ ls not??.txt`	
[ ]	Matches **one character from a specified set**.	`$ ls [a,b,c]*`	
-	Specifies a **range of characters** inside brackets.	`$ ls [a-z]*`	
>	Redirects output to a file, **overwriting existing content**.	`command > file`	
<	Takes input from a file instead of the keyboard.	`command < file`	
|	Sends the output of one command as input to another command.	`command1 | command2`	
$	Accesses the value of a **shell variable**.	`echo $variable`	
cal	Displays a calendar.	`cal [month year]`	
date	Displays the current date and time.	`date`	
cmp	Compares two files **byte by byte**.	`cmp file1 file2`	
comm	Compares **two sorted files line by line**.	`comm file1 file2`	
diff	Shows the differences between two files.	`diff file1 file2`	
head	Displays the first 10 lines by default.	`head [-n N] file`	
tail	Displays the last 10 lines by default.	`tail [-n N] file`	
sort	Sorts lines alphabetically or numerically.	`sort [option] file`	
bc	Performs mathematical calculations.	`echo "expression" | bc`	
expr	Performs arithmetic and string operations.	`expr expression`	
grep	Searches for lines containing a specified pattern.	`grep pattern file`	

1. **Part 1: Exploring Default Permissions and Creating Files**

```
(base) USER@MacBook-Air-3 ~ % mkdir ~/permissions_practice
(base) USER@MacBook-Air-3 ~ % ls -l
drwxr-xr-x    2 USER  staff     64 25 Aug 20:21 permissions_practice
```

```
(base) USER@MacBook-Air-3 ~ % cd ~/permissions_practice
(base) USER@MacBook-Air-3 permissions_practice % touch my_file.txt
```

2. **Part 2: Changing Permissions (Symbolic Mode)**

```
(base) USER@MacBook-Air-3 permissions_practice % touch my_file.txt
(base) USER@MacBook-Air-3 permissions_practice % ls -l
total 0
-rw-r--r--  1 USER  staff  0 25 Aug 20:25 my_file.txt
```

```
(base) USER@MacBook-Air-3 permissions_practice % ls -l my_file.txt
-rw-r--r--  1 USER  staff  0 25 Aug 20:25 my_file.txt
```

```
(base) USER@MacBook-Air-3 permissions_practice % chmod u+x my_file.txt
(base) USER@MacBook-Air-3 permissions_practice % ls -l my_file.txt
-rwxr--r--  1 USER  staff  0 25 Aug 20:25 my_file.txt
(base) USER@MacBook-Air-3 permissions_practice %
```

```
(base) USER@MacBook-Air-3 permissions_practice % chmod g-w my_file.txt
(base) USER@MacBook-Air-3 permissions_practice % ls -l my_file.txt
-rwxr--r--  1 USER  staff  0 25 Aug 20:25 my_file.txt
(base) USER@MacBook-Air-3 permissions_practice %
```

```
(base) USER@MacBook-Air-3 permissions_practice % chmod o=r my_file.txt
(base) USER@MacBook-Air-3 permissions_practice % ls -l my_file.txt
-rwxr--r--  1 USER  staff  0 25 Aug 20:25 my_file.txt
(base) USER@MacBook-Air-3 permissions_practice %
```

3. **Part 3: Changing Permissions (Numeric Mode)**

```
(base) USER@MacBook-Air-3 permissions_practice % touch report1.txt
```

```
(base) USER@MacBook-Air-3 permissions_practice % ls -l report1.txt
-rw-r--r--  1 USER  staff  0 25 Aug 20:36 report1.txt
(base) USER@MacBook-Air-3 permissions_practice % chmod 754 report1.txt
(base) USER@MacBook-Air-3 permissions_practice % ls -l report1.txt
-rwxr-xr--  1 USER  staff  0 25 Aug 20:36 report1.txt
```